

POWERFACTORY

PowerFactory 2024

Technical Reference

Surge Arrester

StaSua, StaSuabi

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

This document describes the models of the surge arrester *StaSua* and the surge arrester with two terminals *StaSuabi*.

In general, the surge arrester model in *PowerFactory* defines a highly nonlinear voltage-current (V-I) characteristic using peak values. This nonlinear characteristic represents the requirement that the surge arrester should have a very high resistance during normal system operation and a low resistance during transient disturbances. The peak voltage-current values are input using the surge arrester element (*StaSua*) via the *Characteristic* table on the *Basic Data* tab of the dialogue.

Values lying between those specifed are interpolated according to the user-selected *Interpolation* mode. For *spline* interpolation, the values are obtained according to the derivative of the first/last point. For *piecewise linear* interpolation, the values are obtained according to the derivative of the first/last inner interval. The interpolated V-I characteristic is then depicted graphically on the *Diagram* tab in the dialogue, as shown in Figure 1.1.

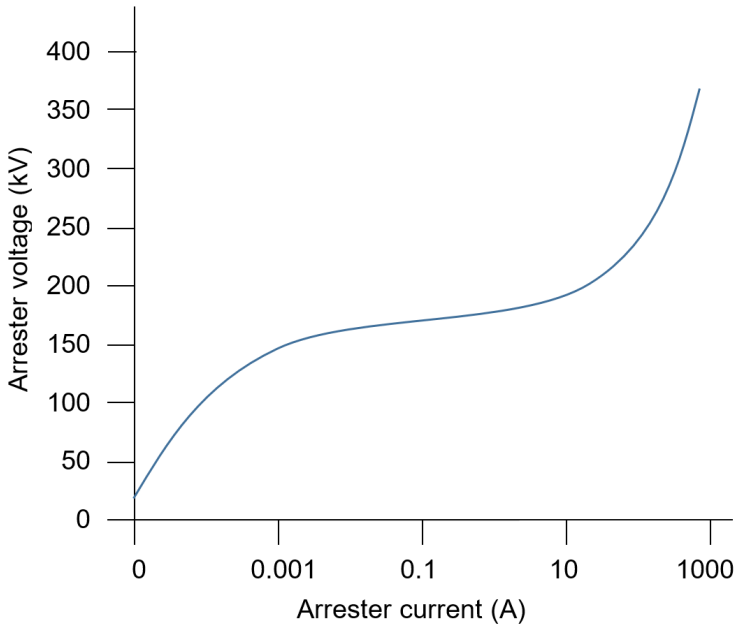


Figure 1.1: Surge arrester characteristic

The characteristic represents the full range of the surge arrester, including its ”normal” condition (i.e. leakage current for the rated voltage). It should be noted that this graphical representation uses a logarithmic scale on the x-axis, and therefore cannot display zero current values entered into the *Characteristic* table.

2 Three phase model

The three phase surge arrester model (consisting of three separate single phase models) can be connected in YN or in D configuration as shown in Figure 2.1.

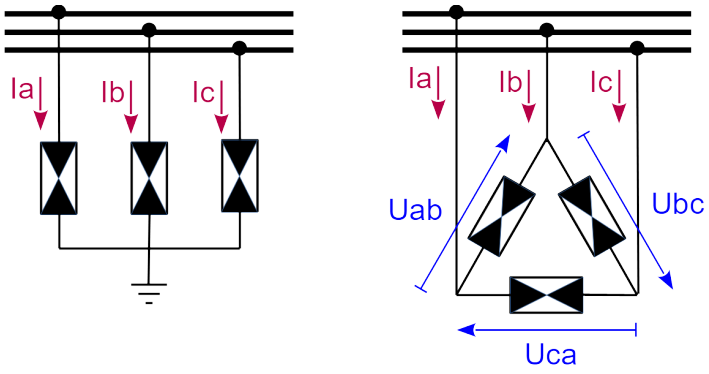


Figure 2.1: Three phase surge arrester connected in YN and D configuration

The model is only for EMT simulation taken into account.

The single phase surge arrester model circuit diagram is shown in Figure 2.2.

The surge arrester is represented as a nonlinear resistance where the peak values of the current (I_{mov}) are interpolated from the current values of the residual voltage (U_{mov}).

The capacitor Cp represents the stray capacitance of the arrester, Ls represents the stray inductance, while Rs represents the resistance (the resistance can be varied using the external signal $inRs$). These values can be estimated based on the physical dimensions of the arrester (which are commonly provided in manufacturers' datasheets), using formulae such as those used in [1].

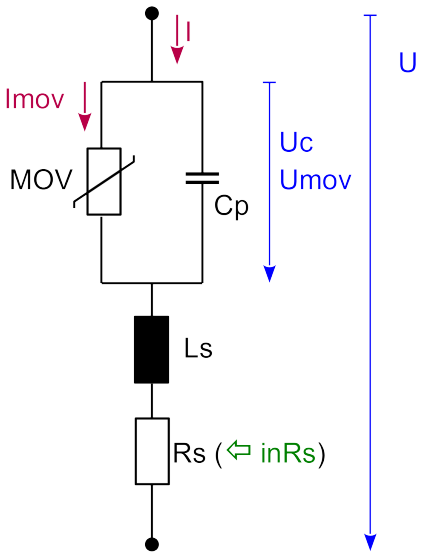


Figure 2.2: Single phase surge arrester model circuit diagram

The surge arrester model has a capacitor, Cp, in parallel with the MOV, leading to the following expression for the phase currents I:

$$I = I_{mov} + Cp \cdot \frac{dU_c}{dt} \tag{1}$$

where dU_c/dt is the derivative of the phase voltage.

The calculation of the voltage at the terminal considers the inductance Ls and the resistance Rs (or the value from $inRs$) which are in series with the MOV:

$$U = U_c + Rs \cdot I + Ls \cdot \frac{dI}{dt} \tag{2}$$

where U is the terminal voltage or voltage difference between the two terminals or phases, U_c is the voltage across the MOV and the capacitor, and dI/dt is the derivative of the phase current.

The instantaneous power is expressed as:

$$P_{mov}(t) = I_{mov}(t) \cdot U_{mov}(t) \tag{3}$$

and the MOV energy is calculated from the instantaneous power as:

$$E_{mov} = \int P_{mov}(t) \tag{4}$$

2.1 Spark Gap

The surge arrester model for EMT simulations allows the definition of a protection spark gap, via the selection of the *Spark Gap* option. Once this option is selected, the resistance and reactance of the damping circuit can be defined. The spark gap model is shown in Figure 2.3.

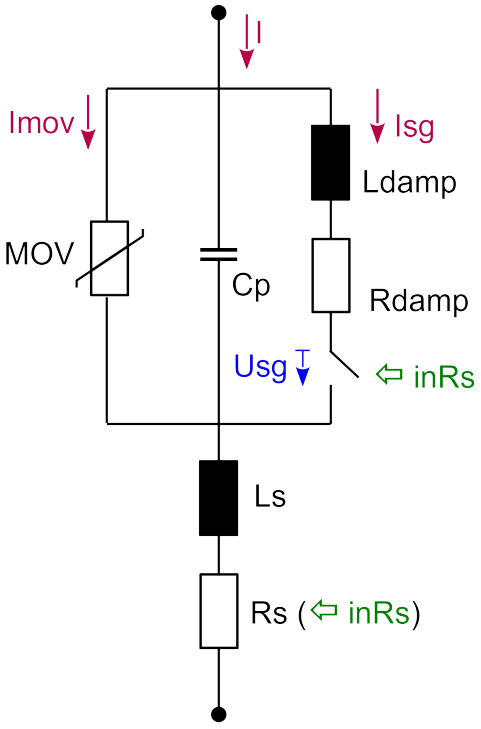


Figure 2.3: Spark gap model

External Tripping When the option *Built-In Tripping Unit* is deactivated, the spark gap can be triggered via the use of input signals *inSg_A*, *inSg_B* and *inSg_C* (for single phase *inSg*) using either a parameter event (*EvtParam*) defined for the EMT simulation, or via an external DSL controller.

Built-In Tripping Unit By activating the option *Built-In Tripping Unit*, it is possible to specify the voltage value at which the spark gap will trigger.

3 Three phase model with two terminals

The three phase surge arrester model with two terminals ((*StaSuabi*)) is shown in Figure 3.1.

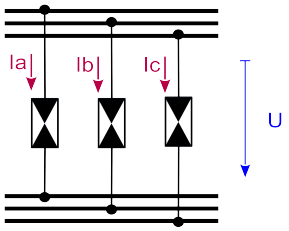


Figure 3.1: Three phase surge arrester with two terminals

The circuit diagram is same as the diagram shown in Figure 2.2 and only taken into account for EMT simulation.
The model described in Section 2 is also valid for the two-terminal model, considering that U is the voltage difference between the connected terminals ($U = Uterminal\ i - Uterminal\ j$).

4 Single phase model (AC or DC)

The single phase surge arrester model circuit diagram is same as the diagram shown in Figure 2.2.

The model is only available for EMT simulation when connected to an AC busbar. In case of a connection at a DC busbar, the mmodel is also for balanced and unbalanced RMS simulation taken into account.

5 Single phase model with two terminals (AC or DC)

The single phase surge arrester model circuit diagram is same as the diagram shown in Figure 2.2.

The model is only available for EMT simulation when connected to an AC busbar. In case of a connection at a DC busbar, the mmodel is also for balanced and unbalanced RMS simulation taken into account.

The model described in Section 2 is also valid for the two-terminal model, considering that U is the voltage difference between the connected terminals ($U = Uterminal\ i - Uterminal\ j$).

A Input parameters, signals and state variables

A.1 Input parameters

Parameter	Unit	Description
Cp	μF	Stray Capacitance
Ls	mH	Series Inductance
Rs	Ω	Series Resistance
iSparkGap		Consider spark gap
Rdamp	Ω	Damping Resistance
Xdamp	Ω	Damping Inductance
iBuiltIn		Use built-in tripping
Uthr	kV	Voltage threshold

Table A.1: Input parameters for the surge arrester

A.2 Signals and state variables

Parameter	Unit	Description
Emov_a	MWs	MOV absorbed energy
Emov_b	MWs	MOV absorbed energy
Emov_c	MWs	MOV absorbed energy
Uc_a	kV	Capacitor voltage
Uc_b	kV	Capacitor voltage
Uc_c	kV	Capacitor voltage
inRs	Ω	Series resistance
inSg_A		Spark gap trigger
inSg_B		Spark gap trigger
inSg_C		Spark gap trigger

Table A.2: Signals for the three phase models

Parameter	Unit	Description
Emov	MWs	MOV absorbed energy
Uc	kV	Capacitor voltage
inRs	Ω	Series resistance
inSg		Spark gap trigger

Table A.3: Signals for the single phase models

Parameter	Unit	Description
Emov	MWs	MOV absorbed energy
Uc	kV	Capacitor voltage

Table A.4: Signals for the DC models

B References

[1] IEEE Working Group 3.4.11. Modeling of Metal Oxide Surge Arresters. *IEEE Transactions on Power Delivery*, 7, 1992.

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